



MIGHTY TRONICS®

« We turn ideas into production-ready hardware »

PUMP INTERFACE FUNCTIONAL & ENGINEERING NOTE

Version & Revision
V 2.0

Michael
info@mightytronics.eu

LIST OF CONTENTS

1.	History of Version	3
2.	purpose and scope	4
2.1.	Why this document?	4
2.2.	Pupose ans scope of MT Drive	4
3.	Declared operating envelope	4
4.	Functional operation.....	5
4.1.	Power-up and idle state	5
4.2.	Command and load energization.....	5
4.3.	Turn-off and inductive-current path	5
4.4.	Protection behaviour	5
5.	External interfaces	6
6.	Functional description	6
7.	Technical implementation.....	6
7.1.	Common power path.....	6
7.2.	Logic and gate-drive path	7
7.3.	Repeated power channel	7
7.4.	Pwm timing	7
8.	Electrical dimensioning	7
8.1.	Published current and load power	7
8.2.	Extended, non-guaranteed region	7
8.3.	MOSFET conduction loss.....	8
8.4.	MOSFET switching loss	8
9.	Current-path and thermal margins.....	8
9.1.	Common reverse-polarity diode.....	8
9.2.	Channel PTC margin	9
9.3.	Copper-path calculation	9
10.	Inductive switching and voltage margin	9
11.	Verification and qualification.....	10
12.	Assembly and first power-up	10
13.	Safety, limitations and use conditions.....	11

14.	Open hardware release package	12
15.	References	12
16.	Appendix A - Released schematic.....	13

1. HISTORY OF VERSION

Date	Version	Amendments
26-09-2026	V1.0	Document initiation.
27-09-2026	V 2.0	Revise doc to adapt for 2A max by channel to ensure safety and quality.

2. PURPOSE AND SCOPE

2.1. WHY THIS DOCUMENT?

This document describes what the four-channel MT Drive board does, how it is connected and how the implemented circuit operates. It then records the component choices, electrical calculations, qualification criteria and publication elements needed to reproduce and maintain the hardware.

2.2. PURPOSE AND SCOPE OF MT DRIVE

MT Drive is a four-channel interface for controlling brushed DC pumps or comparable inductive DC loads from a microcontroller. Each load is switched independently by PWM. The board does not generate the PWM signals, regulate the motor supply or provide galvanic isolation.

This description applies to the schematic titled "Pump Interface", hardware revision V2.0 dated 24 April 2026. The published operating rating is separated from the extended characterization region so that the guaranteed capability remains unambiguous.

3. DECLARED OPERATING ENVELOPE

Published rating: 2 A continuous per channel and 8 A continuous total, from a 5 to 36 V DC motor supply.

Operation up to 3 A per channel may be evaluated, but it is not guaranteed for this hardware revision.

Characteristic	Published specification	Extended characterization
Motor supply	5 to 36 V DC	5 to 36 V DC
Channels	4 independent low-side outputs	Same architecture
Continuous load current	2 A maximum per channel	Up to 3 A per channel, not guaranteed
Aggregate continuous current	8 A maximum	Up to 12 A, not guaranteed
PWM frequency	31.5 kHz nominal	31.5 kHz nominal
Logic supply	+5 V external	+5 V external
PWM input levels	3.3 V or 5 V logic	Same interface
Design ambient	-10 °C to +50 °C	Requires separate characterization

4. FUNCTIONAL OPERATION

Please refer to chapter 16 of this document for the schematic.

4.1. POWER-UP AND IDLE STATE

The motor supply enters through J1. Series diode D5 blocks a reverse-polarity connection. TVS diode D6 and capacitors C1, C2 and C3 limit supply disturbances and provide local energy storage. The protected rail is labelled Vin+ and is available at TP1; TP2 is the power ground reference.

The control electronics receive a separate +5 V supply through J2. The controller ground and the board ground must be connected. Pull-down resistors hold the four PWM inputs and the four MOSFET gates low when the controller is disconnected or its outputs are high-impedance. The default state is therefore all loads off.

4.2.COMMAND AND LOAD ENERGIZATION

A logic-high PWM command is accepted by one section of U1, a 74AHCT125 powered from +5 V. The AHCT input threshold allows a 3.3 V controller signal to be translated to a 5 V logic output. The output then drives the corresponding IRLZ44N gate through the conditioning network.

When the MOSFET is on, current flows from Vin+ through the channel PTC, through the external load connected between Mx+ and Mx–, and through the MOSFET to ground. The load therefore sees the motor-supply voltage during the PWM on-time. The average delivered energy follows the commanded duty cycle.

4.3.TURN-OFF AND INDUCTIVE-CURRENT PATH

When the MOSFET turns off, the current of an inductive load cannot stop instantaneously. The channel flyback diode provides a local recirculation path from Mx– back to Mx+. The resistor 22 Ω + capacitor 1 nF series RC network (snubber) across the output pair damps high-frequency ringing. The input TVS limits residual transients that reach the common supply rail.

4.4.PROTECTION BEHAVIOUR

Each branch includes an MF-R500 resettable fuse. It limits sustained fault energy by increasing resistance as it heats; it is not a precision electronic current limiter. D5 protects against reversed input polarity, D6 clamps supply transients, the flyback diodes control inductive current, and the gate diodes limit negative gate excursions. These functions are complementary.

5. EXTERNAL INTERFACES

Interface	Connection	Function and requirement
J1	PWR, GND	Motor supply input, 5 to 36 V DC. The source and wiring must support the total load current.
J2 pin 1	+5 V	External regulated logic supply for U1; this voltage is not generated on the board.
J2 pin 2	GND	Common reference for the logic supply, PWM source and motor power stage.
J2 pins 3–6	PWM1–PWM4	Independent 3.3 V or 5 V logic commands, 31.5 kHz nominal.
J3 pins 1–8	M3+, M3–, M4+, M4–, M2+, M2–, M1+, M1–	Four two-wire load connections in the physical connector order shown.
TP1 / TP2	Vin+ / GND	Protected motor-rail and ground measurement points.
TP3 / TP10 / TP11 / TP12	Qg1 / Qg2 / Qg3 / Qg4	Gate-waveform test points for channels 1 to 4 respectively.

Integration rules:

- Connect the controller ground to J2 GND before applying PWM signals.
- Apply a regulated +5 V logic supply to J2; do not rely on the motor supply to power U1.
- Use a current-limited motor supply during first power-up and begin with the loads disconnected.
- Size the supply, connector and cable loop for 8 A continuous aggregate current, plus the pump start-current transient.
- Keep PWM and ground wiring short and route motor cables away from sensitive logic or measurement wiring.
- Do not exceed 2 A continuous on any channel or 8 A continuous in total within the published rating.

6. FUNCTIONAL DESCRIPTION

Block	Purpose	Implemented circuit
PF1	Power entry and distribution	J1, D5, D6, C1–C3, Vin+ rail and power-ground return.
PF2	Four load switches	Q1–Q4 with one PTC-protected high-side feed per channel.
PF3	Logic adaptation	U1 74AHCT125 and four input/gate-conditioning networks.
PF4	Inductive-load protection	D1–D4 flyback diodes, R9–R12/C9–C14 snubbers and rail TVS.
PF5	Connection and measurement	J1–J3, TP1–TP3 and TP10–TP12.

7. TECHNICAL IMPLEMENTATION

7.1. COMMON POWER PATH

D5 is a 15SQ100 Schottky diode in series with the positive input. D6 is a 1.5KE43A TVS connected across the protected rail. C1 = 1000 μ F, C2 = 10 μ F and C3 = 100 nF provide bulk, intermediate and high-frequency decoupling. Because D5 carries the sum of all channel currents, its forward drop and thermal path are common-path design limits.

7.2.LOGIC AND GATE-DRIVE PATH

Each PWM input uses a 1 k Ω series resistor and a 100 k Ω pull-down before U1. All four active-low output-enable inputs of the 74AHCT125 are tied low, so the four buffers remain enabled. U1 is decoupled by C34 = 100 nF and C35 = 1 μ F.

Each buffer output drives a 100 Ω gate resistor. The implemented network also includes 1 nF shunt capacitors, a 10 k Ω gate pull-down, a 100 pF gate-source capacitor and a 1N4148 gate clamp. The 74AHCT125 has been bench-tested for the intended 3.3 V-to-5 V signal adaptation. It carries only gate-drive current and does not set the motor-current rating.

7.3.REPEATED POWER CHANNEL

Q1–Q4 are IRLZ44N N-channel MOSFETs used as low-side switches. Each channel contains an MF-R500 PTC in the positive load feed, 22 μ F + 100 nF local rail decoupling, a 15SQ100 flyback diode and a 22 Ω / 0.5 W resistor in series with a 1 nF snubber capacitor across the output pair. The four channels are electrically equivalent apart from their reference designators and connector positions.

7.4.PWM TIMING

$$T = 1 / 31,500 = 31.75 \mu\text{s} \quad \text{PWM period}$$

$$t_{\text{ON}} = 15.87 \mu\text{s at 50\% duty cycle} \quad \text{nominal half-period}$$

The gate waveform must reach stable low and high levels within each PWM period. Duty-cycle preservation is verified at the PWM input, buffer output, MOSFET gate and switched output.

8. ELECTRICAL DIMENSIONING

8.1. PUBLISHED CURRENT AND LOAD POWER

$$I_{\text{total}} = 4 \times 2 \text{ A} = 8 \text{ A}$$

$$P_{\text{channel,max}} = 36 \text{ V} \times 2 \text{ A} = 72 \text{ W}$$

$$P_{\text{load,max}} = 36 \text{ V} \times 8 \text{ A} = 288 \text{ W}$$

The 288 W value is the maximum electrical power delivered to the four external loads. It is not the heat dissipated by the PCB. PCB dissipation is determined by semiconductor voltage drops, conductor resistance, switching losses and connector losses.

8.2.EXTENDED, NON-GUARANTEED REGION

$$I_{\text{total,extended}} = 4 \times 3 \text{ A} = 12 \text{ A}$$

$$P_{\text{load,extended}} = 36 \text{ V} \times 12 \text{ A} = 432 \text{ W}$$

These values define the upper characterization region only. They shall not appear as a **guaranteed product rating**, and operation in this region does not relax the thermal, voltage or protection limits of any component.

8.3.MOSFET CONDUCTION LOSS

The IRLZ44N data sheet specifies a maximum on-state resistance of 46 mΩ at VGS = 4 V and 25 °C. For thermal screening, 80 mΩ is used as a conservative hot-state design value. Full-duty operation is the worst case for conduction loss.

$$P_{\text{cond}} = I^2 \times R_{\text{DS(on)}} \times D \quad D \text{ is the PWM duty cycle}$$

Operating point	RDS(on) basis	Loss per channel	Four-channel total
2 A guaranteed, 25 °C basis	46 mΩ	0.184 W	0.736 W
2 A guaranteed, hot design basis	80 mΩ	0.320 W	1.280 W
3 A extended, 25 °C basis	46 mΩ	0.414 W	1.656 W
3 A extended, hot design basis	80 mΩ	0.720 W	2.880 W

8.4.MOSFET SWITCHING LOSS

$$P_{\text{sw}} = 0.5 \times V_{\text{DS}} \times I_{\text{D}} \times (t_{\text{r}} + t_{\text{f}}) \times f_{\text{PWM}}$$

At 36 V, 2 A and 31.5 kHz, every 100 ns of combined rise and fall time contributes approximately 0.113 W per channel. At 3 A, the same 100 ns contributes approximately 0.170 W per channel. The final value shall use oscilloscope measurements taken directly at the gate and drain on the assembled board.

Current	Switching-loss coefficient	Example at tr + tf = 200 ns	Example at tr + tf = 500 ns
2 A	0.113 W per 100 ns	0.227 W	0.567 W
3 A extended	0.170 W per 100 ns	0.340 W	0.851 W

Total MOSFET loss is the sum of conduction and switching losses. Gate-drive loss, ringing and avalanche energy are assessed separately during prototype validation.

9. CURRENT-PATH AND THERMAL MARGINS

9.1. COMMON REVERSE-POLARITY DIODE

The common 15SQ100 diode carries the aggregate motor current. The manufacturer specifies 15 A average forward current and a maximum forward voltage of 0.82 V at the stated test point. Using 0.82 V provides a conservative thermal screening bound rather than a prediction of the actual operating voltage drop.

Mode	Aggregate current	Screening calculation	Engineering interpretation
Guaranteed	8 A	8 A × 0.82 V = 6.56 W	Thermal qualification required; copper around the diode is a design-critical heat path.
Extended	12 A	12 A × 0.82 V = 9.84 W	High thermal stress; operation is not guaranteed.

The acceptance decision shall use the measured forward drop and stabilized case temperature at 8 A total. The nameplate current alone is not sufficient to establish the board rating.

9.2.CHANNEL PTC MARGIN

Each channel uses an MF-R500 resettable fuse. Its hold current decreases with ambient temperature. The 2 A rating maintains useful margin over the declared $-10\text{ }^{\circ}\text{C}$ to $+50\text{ }^{\circ}\text{C}$ design range. The PTC is a fault-energy limiter; its hold current must not be interpreted as the board's guaranteed load rating.

Ambient temperature	MF-R500 hold current	Margin at 2 A	Margin at 3 A
23 °C	5.00 A	2.50×	1.67×
40 °C	4.15 A	2.08×	1.38×
50 °C	3.85 A	1.93×	1.28×

9.3.COPPER-PATH CALCULATION

For a representative 100 mm conductor segment, 2.0 mm wide and 70 μm thick, the calculated resistance at 20 °C is 12.3 m Ω . This normalized segment is used to compare current levels; release approval shall be based on the actual CAD current paths, neck-downs, vias and connector footprints.

Current through segment	Voltage drop	Copper loss	Use
2 A	24.6 mV	0.049 W	Guaranteed branch current
3 A	36.9 mV	0.111 W	Extended branch current
8 A	98.4 mV	0.787 W	Guaranteed aggregate current if geometry were identical
12 A	147.6 mV	1.771 W	Extended aggregate current if geometry were identical

The common power path must use substantially more copper than a single branch and shall be checked as a complete loop, including the return path.

10.INDUCTIVE SWITCHING AND VOLTAGE MARGIN

The flyback diode, RC snubber and supply TVS have distinct functions. The flyback diode maintains motor-current continuity, the 22 Ω + 1 nF snubber damps local ringing, and the TVS limits supply-rail transients. None of these functions replaces the others.

At a 36 V supply, the 55 V IRLZ44N has 19 V of absolute drain-voltage headroom. The internal reliability target is therefore a measured repetitive VDS peak no greater than 45 V under the guaranteed load, longest intended cable and worst switching condition. Any result above this target requires snubber adjustment, loop-layout correction or a higher-voltage MOSFET in the next revision.

The implemented snubber value is 22 Ω + 1 nF. It shall be confirmed on the assembled board from measured waveforms at 31.5 kHz and with the longest intended motor cable.

11. VERIFICATION AND QUALIFICATION

The guaranteed rating becomes releasable only after the following evidence is recorded on the production-representative prototype.

ID	Verification	Test condition	Acceptance criterion
QV-01	Four-channel logic	3.3 V and 5 V input; 31.5 kHz; 10%, 50%, 90% duty	All channels reproduce logic levels and duty cycle without abnormal distortion.
QV-02	Single-channel power	36 V, 2 A, 0% to 100% duty	Stable operation; no unintended turn-on or oscillation.
QV-03	Full guaranteed load	36 V, 2 A per channel, four channels, two-hour thermal stabilization	8 A continuous operation without protection trip or degradation.
QV-04	High ambient	Guaranteed load at 50 °C ambient	Power-component case temperatures remain within the 90 °C internal target.
QV-05	Switching stress	Longest intended harness; synchronized and phase-shifted PWM	Repetitive VDS peak $\leq$ 45 V; no uncontrolled ringing.
QV-06	Start and stall	Representative pumps and supply	No permanent damage; currents and PTC behaviour documented.
QV-07	Endurance	24 h at guaranteed envelope with duty-cycle changes	No drift, damage, connector discoloration or functional interruption.
QV-08	Extended characterization	Progressive tests above 2 A up to 3 A per channel	Results documented as engineering data only; no guarantee created.
QV-09	EMC pre-compliance	Final PCB, pumps, supply and cables	Emissions and immunity results recorded against the project test plan.

12.ASSEMBLY AND FIRST POWER-UP

- Verify part orientation for D1–D6, Q1–Q4 and polarized capacitors before applying power.
- Inspect all SMD and through-hole solder joints on the released PCB and check that no adjacent gate or output nets are bridged.
- With power removed, check resistance from Vin+ to GND and from each Mx–output to GND. A continuity-tester beep alone is not a short-circuit diagnosis because semiconductor junctions and pull-down networks create measurable paths.
- Apply +5 V logic power first with a current limit and no motor supply. Confirm U1 supply, low idle gates and a clean 0–5 V gate response on all four channels.
- Apply a low motor voltage with one current-limited resistive load, then repeat channel by channel before connecting pumps.
- Increase voltage and current progressively while recording D5, PTC, MOSFET and connector temperatures.

13.SAFETY, LIMITATIONS AND USE CONDITIONS

- The board is a component for integration, not a complete protected end product.
- There is no galvanic isolation between logic ground and motor-power ground.
- The board does not regulate motor voltage and does not actively limit channel current.
- The integrating system must provide a correctly rated external supply, wiring protection, enclosure, strain relief and emergency-disconnect strategy where required.
- Pump stall and start currents must be verified against the 2 A continuous rating and the measured transient response.
- Operation up to 3 A per channel is characterization only and creates no guaranteed operating entitlement.

14.OPEN HARDWARE RELEASE PACKAGE

This document supports an open hardware release but does not by itself establish OSHWA certification. The public repository must identify the exact hardware version and provide the editable design sources needed to study, modify and reproduce it.

Repository content	Minimum published material	Release status
Hardware source	Native KiCad project, schematic and PCB layout files for the released version.	Required
Manufacturing	Gerbers, drill files, fabrication notes, assembly drawings and pick-and-place data where applicable.	Required for reproduction
Bill of materials	Reference, quantity, value, package, manufacturer and orderable part number.	Required
Documentation	This description, schematic PDF, connector pinout, assembly guidance and validation records.	Required
License	An OSHWA-compatible hardware license applied to the hardware sources; an open license applied to documentation.	Select before certification
Version traceability	PCB marking or release date matched to an immutable repository tag and its documentation.	Required
Certification mark	Use only after OSHWA approval and assignment of a project UID.	Not applicable before approval

15.REFERENCES

1. Mighty Tronics, Pump Interface schematic, hardware revision V2.0, 24 April 2026.
2. Infineon Technologies, IRLZ44N HEXFET Power MOSFET data sheet.
3. Bourns, MF-R Series PTC Resettable Fuses data sheet.
4. Rectron, 15SQ100 Schottky Barrier Rectifier data sheet.
5. Nexperia, 74AHC125 and 74AHCT125 Quad Buffer and Line Driver data sheet.
6. Open Source Hardware Association, Open Source Hardware Definition, <https://oshwa.org/definition/>.
7. Open Source Hardware Association, Certification Requirements, <https://certification.oshwa.org/requirements.html>.
8. Open Source Hardware Association, Certification Process — Documentation and Hardware, <https://certification.oshwa.org/process/>.

Project name: Pump Interface
Version : 2.0

